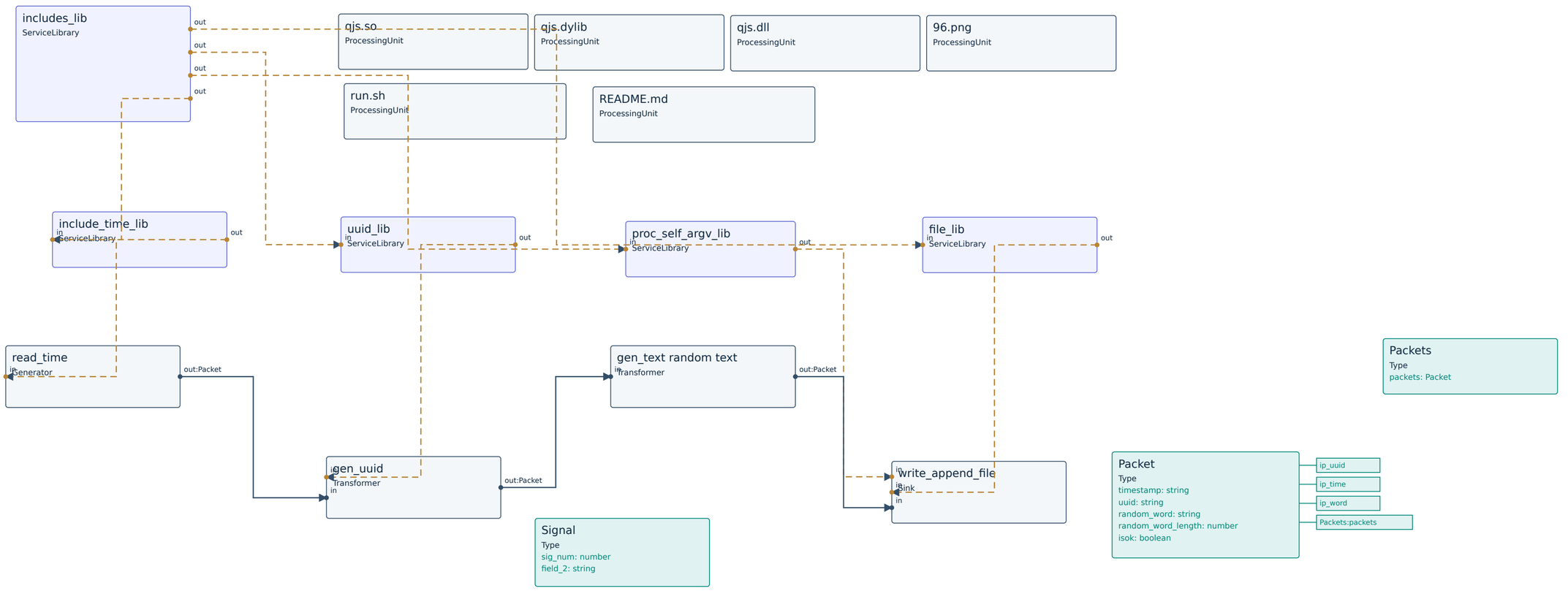
ticker

PARTS LIST

No.	Part	Kind	Modified	Rev.	Responsible	Signature
1	96.png	ProcessingUnit	2026-09-17T01:34:32.418280335Z	A	devlin	
2	file_lib	ServiceLibrary	2026-09-17T01:35:38.882664409Z	A	devlin	
3	gen_text	Transformer	2026-09-17T01:35:37.798459440Z	A	devlin	
4	gen_uuid	Transformer	2026-09-17T01:37:27.489014179Z	A	devlin	
5	include_time_lib	ServiceLibrary	2026-09-17T01:35:40.159634913Z	A	devlin	
6	includes_lib	ServiceLibrary	2026-09-17T01:35:42.882429852Z	A	devlin	
7	Packet	Type	2026-09-17T01:35:19.430086813Z	A	devlin	
8	Packets	Type	2026-09-17T01:44:55.501011488Z	A	devlin	
9	proc_self_argv_lib	ServiceLibrary	2026-09-17T01:35:30.097985811Z	A	devlin	
10	qjs.dll	ProcessingUnit	2026-09-17T01:34:52.393906019Z	A	devlin	
11	qjs.dylib	ProcessingUnit	2026-09-17T01:34:54.414705509Z	A	devlin	
12	qjs.so	ProcessingUnit	2026-09-17T01:34:56.234390118Z	A	devlin	
13	read_time	Generator	2026-09-17T01:35:34.489267648Z	A	devlin	
14	README.md	ProcessingUnit	2026-09-17T01:35:00.932809344Z	A	devlin	
15	run.sh	ProcessingUnit	2026-09-17T01:35:03.365282309Z	A	devlin	
16	Signal	Type	2026-09-17T01:35:23.560607851Z	A	devlin	
17	uuid_lib	ServiceLibrary	2026-09-17T01:35:31.244526113Z	A	devlin	
18	write_append_file	Sink	2026-09-17T01:35:35.662104229Z	A	devlin	



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Revision: A · Wirecraft technical plan

ticker

ticker

Revision: A · 20260831

Format: A2-landscape · Scale 1:1 · Sheet 1/1 · Wirecraft

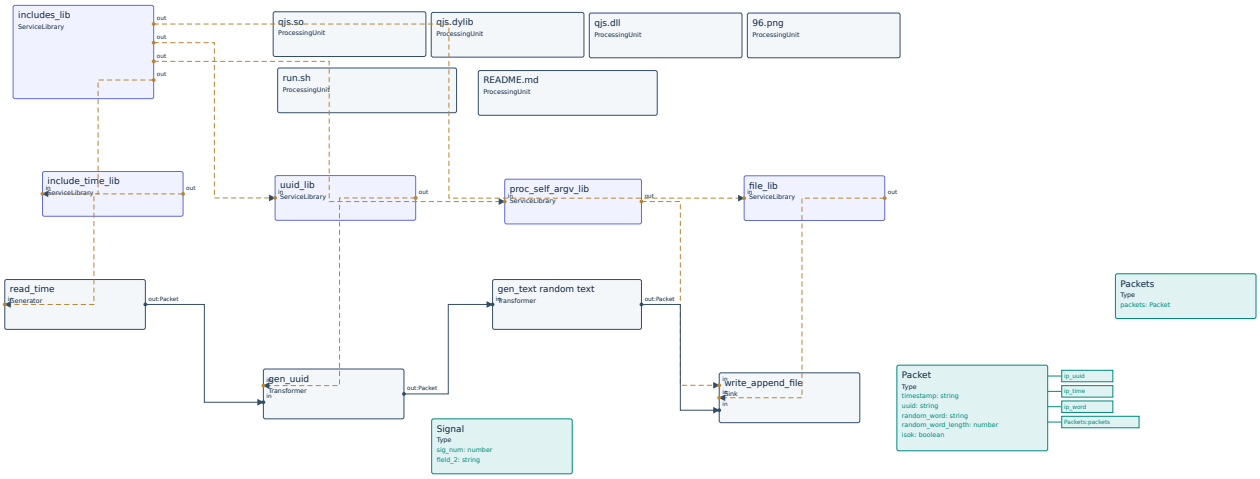
ticker Functional and Technical Specification

Contents

- ticker
- ticker/run.sh
- ticker/Packet
- ticker/file_lib
- ticker/files
- ticker/includes5
- ticker/gen_text
- ticker/ip_word
- ticker/ip_uuid
- ticker/gen_uuid
- ticker/ip_time
- ticker/uuids
- ticker/include_time_lib
- ticker/tm
- ticker/includes3
- ticker/includes_lib
- ticker/includes2
- ticker/includes6
- ticker/proc_self_argv_lib
- ticker/argslib
- ticker/read_time
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- ticker/write_append_file
- ticker/README.md
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- ticker/qjs.dll
- ticker/qjs.dylib
- ticker/qjs.so
- ticker/Signal
- ticker/Packets

Composition · ticker

ticker



ticker

- ID: `root`
- Kind: Processor
- Role: CompositeWorker
- Path: ticker
- Language: c
- Technology: c-native
- Status:
- Assignee:
- Responsible: devlin

Specification

this workflow;
 - reads the system time
 - then packs it up with an uuid
 - then adds a random word
 - then finishes by adding the structure the structure to a file

run.sh

- ID: `node_968f03ce-27b7-4776-a435-ae0d3a01085a`
- Kind: Processor
- Role: ProcessingUnit
- Path: ticker/run.sh
- Language: shellscript
- Technology: file-export
- Status: DEPLOYED
- Assignee:
- Responsible: devlin

Source

```
tcc ticker.c -Wall -luuid -o ticker
./ticker sleep 1000000
```

variant using the embedded tcc compiler

```
java -jar threadworks.jar tinycc --launcher java exe ticker.c -luuid -o ticker
javac -cp ../lib/tinycc-embed.jar TickerjLauncher.java
java -cp .:threadworks.jar TickerjLauncher --sleep 1000000
```

Packet

- ID: `node_c4fb01af-8d9d-4a30-838f-8ab8f6328db8`
- Kind: Type
- Role: Type
- Path: ticker/Packet
- Language: c
- Technology: c-native
- Status: DEPLOYED
- Assignee:

- Responsible: devlin

file_lib

- ID: `node_b5041d57-c1e9-4f67-9e88-760c35c8efde`
- Kind: Processor
- Role: ServiceLibrary
- Path: ticker/file_lib
- Language: c
- Technology: c-native
- Status: TESTING
- Assignee:
- Responsible: devlin

Specification

includes stdio.h or whatever is necessary for the source using the lib to have access to fopen write, fclose et al

Source

```
#ifndef DEF_THREADWORK_ERROR
#define THREADWORK_ERROR -1
#endif
```

- Link `link_1db11e5a-fdb1-4f8b-9109-738451e3584a`: node_b886239f-e4d3-490b-be5d-4e0d0792565b:out → node_b5041d57-c1e9-4f67-9e88-760c35c8efde:in (lib,)
- Link `link_3f327a51-135b-4216-b55f-70ae41f06a30`: node_b5041d57-c1e9-4f67-9e88-760c35c8efde:out → node_3b600a6f-96a5-42b4-beae-b74338ccf170:in (lib,)

files

- ID: `link_3f327a51-135b-4216-b55f-70ae41f06a30`
- Kind: Link
- Role: Link
- Path: ticker/files
- Language: c
- Technology: c-native
- Status:
- Assignee:
- Responsible: devlin

Link contract


```
{
  "sourceNodeId": "node_b5041d57-c1e9-4f67-9e88-760c35c8efde",
  "sourcePortName": "out",
  "targetNodeId": "node_3b600a6f-96a5-42b4-beae-b74338ccf170",
  "targetPortName": "in",
  "transportKind": "in-process",
  "interactionKind": "lib",
  "typeName": "",
```

```
  "payloadDefinition": "",
  "typeDefinitionId": "",
  "compositeBoundaryIds": [],
  "typeExpression": null
}
```


includes5

- ID: `link_1db11e5a-fdb1-4f8b-9109-738451e3584a`
- Kind: Link
- Role: Link
- Path: ticker/includes5
- Language: c
- Technology: c-native
- Status:
- Assignee:
- Responsible: devlin

Link contract


```
{
  "sourceNodeId": "node_b886239f-e4d3-490b-be5d-4e0d0792565b",
  "sourcePortName": "out",
  "targetNodeId": "node_b5041d57-c1e9-4f67-9e88-760c35c8efde",
  "targetPortName": "in",
  "transportKind": "in-process",
  "interactionKind": "lib",
  "typeName": "",
  "payloadDefinition": "",
  "typeDefinitionId": "",
  "compositeBoundaryIds": [],
  "typeExpression": null
}
```


gen_text

- ID: `node_059814d5-2184-47b6-93b0-f754589cf34f`
- Kind: Processor
- Role: Transformer
- Path: ticker/gen_text
- Language: c
- Technology: c-native
- Status: BUSINESS
- Assignee:
- Responsible: devlin

Specification

```
takes a Packet from ip_uuid and generates a random word string of length random_word_length using the diction
aries

consonants="bcdfgklmnprstuvxz"
```

```
vowels="aeiou"

alternating consonants with vowels and starting with a consonant

finally it pushes the updated Packet to ip_word
```

Source

```
static const char consonants[] = "bcdfgklmnprstuvxz";
static const char vowels[] = "aeiou";

Packet item;
if (pop(ip_uuid, &item) != THREADWORK_OK) {
    return THREADWORK_OK;
}


// should raise a compilation error // azurty;

int length = (int)item.random_word_length;

if (length < 0) {
    item.isok = false;
    push(ip_word, &item);
    return THREADWORK_ERROR;
}

char *word = malloc((size_t)length + 1);

if (word == NULL) {
    item.isok = false;
    push(ip_word, &item);
    return THREADWORK_ERROR;
}

// should raise another compilation error // azurty;
for (int i = 0; i < length; ++i) {
    if ((i % 2) == 0) {
        word[i] = consonants[
            rand() % (sizeof(consonants) - 1)
        ];
    } else {
        word[i] = vowels[
            rand() % (sizeof(vowels) - 1)
        ];
    }
}

// shouldnt raise another compilation error because tcc is skibidy // azurty;
word[length] = '\0';

item.random_word = word;
printf("gen_text.timestamp:%s\n",item.timestamp != NULL ? item.timestamp : "null");
printf("gen_text.uuid:%s\n",item.uuid != NULL ? item.uuid : "null");
```

```
printf("gen_text.random_word:%s\n",item.random_word != NULL ? item.random_word : "null");

push(ip_word, &item);
```

- Link `link_8e881876-8d2a-41a8-92a4-2ee3f2c839ce`: node_627a3a3e-718f-4867-a71e-584b2d9b138b:out → node_059814d5-2184-47b6-93b0-f754589cf34f:in (data,)
- Link `link_dd5fcd39-8d54-4310-b6fe-242d927f0ec5`: node_059814d5-2184-47b6-93b0-f754589cf34f:out → node_3b600a6f-96a5-42b4-beae-b74338ccf170:in (data,)

ip_word

- ID: `link_dd5fcd39-8d54-4310-b6fe-242d927f0ec5`
- Kind: Link
- Role: Link
- Path: ticker/ip_word
- Language: c
- Technology: c-native
- Status:
- Assignee:
- Responsible: devlin

Link contract


```
{
  "sourceNodeId": "node_059814d5-2184-47b6-93b0-f754589cf34f",
  "sourcePortName": "out",
  "targetNodeId": "node_3b600a6f-96a5-42b4-beae-b74338ccf170",
  "targetPortName": "in",
  "transportKind": "in-process",
  "interactionKind": "data",
  "typeName": "",
  "payloadDefinition": "",
  "typeDefinitionId": "node_c4fb01af-8d9d-4a30-838f-8ab8f6328db8",
  "compositeBoundaryIds": [],
  "typeExpression": null
}
```


ip_uuid

- ID: `link_8e881876-8d2a-41a8-92a4-2ee3f2c839ce`
- Kind: Link
- Role: Link
- Path: ticker/ip_uuid
- Language: c
- Technology: c-native
- Status:
- Assignee:
- Responsible: devlin

Link contract


```
{
  "sourceNodeId": "node_627a3a3e-718f-4867-a71e-584b2d9b138b",
```

```
  "sourcePortName": "out",
  "targetNodeId": "node_059814d5-2184-47b6-93b0-f754589cf34f",
  "targetPortName": "in",
  "transportKind": "in-process",
  "interactionKind": "data",
  "typeName": "",
  "payloadDefinition": "",
  "typeDefinitionId": "node_c4fb01af-8d9d-4a30-838f-8ab8f6328db8",
  "compositeBoundaryIds": [],
  "typeExpression": null
}
```


gen_uuid

- ID: `node_627a3a3e-718f-4867-a71e-584b2d9b138b`
- Kind: Processor
- Role: Transformer
- Path: ticker/gen_uuid
- Language: c
- Technology: c-native
- Status: FUNCTIONAL
- Assignee:
- Responsible: devlin

Specification

takes a Packet from ip_time, fills in its uuid field, sets random_word_length to a number between 5 and 25, and pushes the updated packet to ip_uuid

Source

```
Packet item;
if (pop(ip_time, &item) != THREADWORK_OK) {
    return THREADWORK_OK;
}

// should raise a compilation error // -gibberish--

uuid_t uuid_value;
char *uuid_string = malloc(37);

if (uuid_string == NULL) {
    item.isok = false;
    push(ip_uuid, &item);
    return THREADWORK_ERROR;
}

uuids__uuid_generate_random(uuid_value);
uuids__uuid_unparse_lower(uuid_value, uuid_string);

item.uuid = uuid_string;
item.random_word_length = 5 + (uuid_value[0] % 21);
```

```
printf("gen_text.timestamp:%s\n",item.timestamp != NULL ? item.timestamp : "null");
printf("gen_text.uuid:%s\n",item.uuid != NULL ? item.uuid : "null");

push(ip_uuid, &item);

pu
```

- Link `link_a17dca3c-a162-4f7f-a264-abcd20c1f421`: node_b153ded5-bb9d-438c-9854-84fa0e737261:out → node_627a3a3e-718f-4867-a71e-584b2d9b138b:in (lib,)
- Link `link_d1b9d286-d43a-439a-bee0-e66bf2943400`: node_c43e7a55-0e44-428b-972a-6dab999df396:out → node_627a3a3e-718f-4867-a71e-584b2d9b138b:in (data,)
- Link `link_8e881876-8d2a-41a8-92a4-2ee3f2c839ce`: node_627a3a3e-718f-4867-a71e-584b2d9b138b:out → node_059814d5-2184-47b6-93b0-f754589cf34f:in (data,)

ip_time

- ID: `link_d1b9d286-d43a-439a-bee0-e66bf2943400`
- Kind: Link
- Role: Link
- Path: ticker/ip_time
- Language: c
- Technology: c-native
- Status:
- Assignee:
- Responsible: devlin

Link contract


```
{
  "sourceNodeId": "node_c43e7a55-0e44-428b-972a-6dab999df396",
  "sourcePortName": "out",
  "targetNodeId": "node_627a3a3e-718f-4867-a71e-584b2d9b138b",
  "targetPortName": "in",
  "transportKind": "in-process",
  "interactionKind": "data",
  "typeName": "",
  "payloadDefinition": "",
  "typeDefinitionId": "node_c4fb01af-8d9d-4a30-838f-8ab8f6328db8",
  "compositeBoundaryIds": [],
  "typeExpression": null
}
```


uuids

- ID: `link_a17dca3c-a162-4f7f-a264-abcd20c1f421`
- Kind: Link
- Role: Link
- Path: ticker/uuids
- Language: c
- Technology: c-native
- Status:
- Assignee:
- Responsible: devlin

Link contract


```
{
  "sourceNodeId": "node_b153ded5-bb9d-438c-9854-84fa0e737261",
  "sourcePortName": "out",
  "targetNodeId": "node_627a3a3e-718f-4867-a71e-584b2d9b138b",
  "targetPortName": "in",
  "transportKind": "in-process",
  "interactionKind": "lib",
  "typeName": "",
  "payloadDefinition": "",
  "typeDefinitionId": "",
  "compositeBoundaryIds": [],
  "typeExpression": null
}
```


include_time_lib

- ID: `node_82203592-4fed-4fd4-b0bf-9b02a90b0ad0`
- Kind: Processor
- Role: ServiceLibrary
- Path: ticker/include_time_lib
- Language: c
- Technology: c-native
- Status: TESTING
- Assignee:
- Responsible: devlin

Specification

creates a function that gets the system time and returns it as an int

Source

```c
#ifndef TIME_LIB_H
#define TIME_LIB_H

#include <stdbool.h>
#include <stdio.h>
#include <time.h>

#endif /* TIME_LIB_H */
```

- Link `link_cb8f1b19-7f21-432b-8055-3340b5513424`: node_b886239f-e4d3-490b-be5d-4e0d0792565b:out → node_82203592-4fed-4fd4-b0bf-9b02a90b0ad0:in (lib,)
- Link `link_ecc22fe5-032c-4b24-812b-cc93b4ca76b0`: node_82203592-4fed-4fd4-b0bf-9b02a90b0ad0:out → node_c43e7a55-0e44-428b-972a-6dab999df396:in (lib,)

tm

- ID: `link_ecc22fe5-032c-4b24-812b-cc93b4ca76b0`

- Kind: Link
- Role: Link
- Path: ticker/tm
- Language: c
- Technology: c-native
- Status:
- Assignee:
- Responsible: devlin

Link contract


```
{
  "sourceNodeId": "node_82203592-4fed-4fd4-b0bf-9b02a90b0ad0",
  "sourcePortName": "out",
  "targetNodeId": "node_c43e7a55-0e44-428b-972a-6dab999df396",
  "targetPortName": "in",
  "transportKind": "in-process",
  "interactionKind": "lib",
  "typeName": "",
  "payloadDefinition": "",
  "typeDefinitionId": "",
  "compositeBoundaryIds": [],
  "typeExpression": null
}
```


includes3

- ID: `link_cb8f1b19-7f21-432b-8055-3340b5513424`
- Kind: Link
- Role: Link
- Path: ticker/includes3
- Language: c
- Technology: c-native
- Status:
- Assignee:
- Responsible: devlin

Link contract


```
{
  "sourceNodeId": "node_b886239f-e4d3-490b-be5d-4e0d0792565b",
  "sourcePortName": "out",
  "targetNodeId": "node_82203592-4fed-4fd4-b0bf-9b02a90b0ad0",
  "targetPortName": "in",
  "transportKind": "in-process",
  "interactionKind": "lib",
  "typeName": "",
  "payloadDefinition": "",
  "typeDefinitionId": "",
  "compositeBoundaryIds": [],
  "typeExpression": null
}
```

includes_lib

- ID: `node_b886239f-e4d3-490b-be5d-4e0d0792565b`
- Kind: Processor
- Role: ServiceLibrary
- Path: ticker/includes_lib
- Language: c
- Technology: c-native
- Status: TESTING
- Assignee:
- Responsible: devlin

Specification

```
#include declarations
```

Source

```
#ifndef INCLUDES_LIB_H
#define INCLUDES_LIB_H

#include <stdbool.h>
#include <stdio.h>
#include <time.h>
#include <string.h>
#include <stdbool.h>
#include <uuid/uuid.h>
#include <stdio.h>
#include <stdio.h>
#include <signal.h>
#include <unistd.h>

#endif /* INCLUDES_LIB_H */
```

- Link `link_1db11e5a-fdb1-4f8b-9109-738451e3584a`: node_b886239f-e4d3-490b-be5d-4e0d0792565b:out → node_b5041d57-c1e9-4f67-9e88-760c35c8efde:in (lib,)
- Link `link_2f93c7bf-0dcc-4589-8e80-ad66d57b3862`: node_b886239f-e4d3-490b-be5d-4e0d0792565b:out → node_b153ded5-bb9d-438c-9854-84fa0e737261:in (lib,)
- Link `link_6d0edf8c-7c32-452a-ab0d-dd384f8cfffc`: node_b886239f-e4d3-490b-be5d-4e0d0792565b:out → node_98b1de57-6b1a-4a96-8e55-6e9a11eb1fc4:in (lib,)
- Link `link_cb8f1b19-7f21-432b-8055-3340b5513424`: node_b886239f-e4d3-490b-be5d-4e0d0792565b:out → node_82203592-4fed-4fd4-b0bf-9b02a90b0ad0:in (lib,)

includes2

- ID: `link_6d0edf8c-7c32-452a-ab0d-dd384f8cfffc`
- Kind: Link
- Role: Link
- Path: ticker/includes2
- Language: c
- Technology: c-native
- Status:

- Assignee:
- Responsible: devlin

Link contract


```
{
  "sourceNodeId": "node_b886239f-e4d3-490b-be5d-4e0d0792565b",
  "sourcePortName": "out",
  "targetNodeId": "node_98b1de57-6b1a-4a96-8e55-6e9a11eb1fc4",
  "targetPortName": "in",
  "transportKind": "in-process",
  "interactionKind": "lib",
  "typeName": "",
  "payloadDefinition": "",
  "typeDefinitionId": "",
  "compositeBoundaryIds": [],
  "typeExpression": null
}
```


includes6

- ID: `link_2f93c7bf-0dcc-4589-8e80-ad66d57b3862`
- Kind: Link
- Role: Link
- Path: ticker/includes6
- Language: c
- Technology: c-native
- Status:
- Assignee:
- Responsible: devlin

Link contract


```
{
  "sourceNodeId": "node_b886239f-e4d3-490b-be5d-4e0d0792565b",
  "sourcePortName": "out",
  "targetNodeId": "node_b153ded5-bb9d-438c-9854-84fa0e737261",
  "targetPortName": "in",
  "transportKind": "in-process",
  "interactionKind": "lib",
  "typeName": "",
  "payloadDefinition": "",
  "typeDefinitionId": "",
  "compositeBoundaryIds": [],
  "typeExpression": null
}
```


proc_self_argv_lib

- ID: `node_98b1de57-6b1a-4a96-8e55-6e9a11eb1fc4`
- Kind: Processor
- Role: ServiceLibrary

- Path: ticker/proc_self_argv_lib
- Language: c
- Technology: c-native
- Status: IMPLEMENTATION
- Assignee:
- Responsible: devlin

Specification

reads the commandline from "/proc/self/cmdline"

Source

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#define ARGPARSE_NOTERM  -1
#define ARGPARSE_NOVALUE -2
#define ARGPARSE_OOBOUND -3

static char  *argparse_buffer = NULL;
static char **argparse_argv   = NULL;
static int    argparse_argc   = 0;
static int    argparse_loaded = 0;


/* ---------------------------------------------------------- */
/* internal loading                                           */
/* ---------------------------------------------------------- */

static int argparse_load(void)
{

    if (argparse_loaded)
        return 0;

    FILE *f = fopen("/proc/self/cmdline", "rb");
    if (f == NULL)
        return ARGPARSE_NOTERM;

    size_t capacity = 256;
    size_t size = 0;

    argparse_buffer = malloc(capacity);
    if (argparse_buffer == NULL) {
        fclose(f);
        return ARGPARSE_OOBOUND;
    }

    for (;;) {
        if (size == capacity) {
            capacity *= 2;
```

```
            char *new_buffer = realloc(argparse_buffer, capacity);
            if (new_buffer == NULL) {
                free(argparse_buffer);
                argparse_buffer = NULL;
                fclose(f);
                return ARGPARSE_OOBOUND;
            }

            argparse_buffer = new_buffer;
        }

        size_t n = fread(
            argparse_buffer + size,
            1,
            capacity - size,
            f
        );

        size += n;

        if (n == 0)
            break;
    }

    fclose(f);

    if (size == 0)
        return ARGPARSE_NOTERM;

    /*
     * /proc/self/cmdline normally ends in \0,
     * but guarantee it ourselves.
     */
    if (argparse_buffer[size - 1] != '\0') {
        if (size == capacity) {
            capacity++;

            char *new_buffer = realloc(argparse_buffer, capacity);
            if (new_buffer == NULL)
                return ARGPARSE_OOBOUND;

            argparse_buffer = new_buffer;
        }

        argparse_buffer[size++] = '\0';
    }

    /* count arguments */
    for (size_t i = 0; i < size; i++) {
        if (argparse_buffer[i] == '\0')
            argparse_argc++;
    }

    argparse_argv = malloc(
        sizeof(char *) * (size_t)argparse_argc
```

```
    );

    if (argparse_argv == NULL)
        return ARGPARSE_OOBOUND;

    int index = 0;
    char *p = argparse_buffer;
    char *end = argparse_buffer + size;

    while (p < end && index < argparse_argc) {
        argparse_argv[index++] = p;
        p += strlen(p) + 1;
    }

    argparse_loaded = 1;
    return 0;
}


/* ---------------------------------------------------------- */
/* helpers                                                    */
/* ---------------------------------------------------------- */

static int option_like(const char *s)
{
    return s != NULL &&
           s[0] == '-' &&
           s[1] != '\0';
}


/* ---------------------------------------------------------- */
/* --name value                                               */
/* ---------------------------------------------------------- */

int option(const char *name, char **option_value)
{
    if (argparse_load() < 0)
        return ARGPARSE_NOTERM;

    if (name == NULL || option_value == NULL)
        return ARGPARSE_NOTERM;

    *option_value = NULL;

    for (int i = 1; i < argparse_argc; i++) {

        const char *arg = argparse_argv[i];

        if (strncmp(arg, "--", 2) != 0)
            continue;

        const char *actual_name = arg + 2;

        if (strcmp(actual_name, name) == 0) {
```

```
            if (i + 1 >= argparse_argc)
                return ARGPARSE_NOVALUE;

            if (option_like(argparse_argv[i + 1]))
                return ARGPARSE_NOVALUE;

            *option_value = argparse_argv[i + 1];
            return i;
        }
    }

    return ARGPARSE_NOTERM;
}


/* ------------------------------------------------------- */
/* -flag                                                   */
/* ------------------------------------------------------- */

int flag(const char *name)
{
    if (argparse_load() < 0)
        return ARGPARSE_NOTERM;

    if (name == NULL)
        return ARGPARSE_NOTERM;

    for (int i = 1; i < argparse_argc; i++) {

        const char *arg = argparse_argv[i];

        if (arg[0] != '-' || arg[1] == '-')
            continue;

        if (strcmp(arg + 1, name) == 0)
            return i;
    }

    return ARGPARSE_NOTERM;
}


/* ------------------------------------------------------- */
/* -p value                                                */
/* ------------------------------------------------------- */

int short_option(const char *name, char **option_value)
{
    if (argparse_load() < 0)
        return ARGPARSE_NOTERM;

    if (name == NULL || option_value == NULL)
        return ARGPARSE_NOTERM;

    *option_value = NULL;
```

```
    for (int i = 1; i < argparse_argc; i++) {

        const char *arg = argparse_argv[i];

        if (arg[0] != '-' || arg[1] == '-')
            continue;

        if (strcmp(arg + 1, name) == 0) {

            if (i + 1 >= argparse_argc)
                return ARGPARSE_NOVALUE;

            if (option_like(argparse_argv[i + 1]))
                return ARGPARSE_NOVALUE;

            *option_value = argparse_argv[i + 1];
            return i;
        }
    }

    return ARGPARSE_NOTERM;
}


/* ---------------------------------------------------------- */
/* literal word                                               */
/* ---------------------------------------------------------- */

int word(const char *name)
{
    if (argparse_load() < 0)
        return ARGPARSE_NOTERM;

    if (name == NULL)
        return ARGPARSE_NOTERM;

    for (int i = 1; i < argparse_argc; i++) {

        if (strcmp(argparse_argv[i], name) == 0)
            return i;
    }

    return ARGPARSE_NOTERM;
}


/* ---------------------------------------------------------- */
/* everything from argv[startpos] onward                      */
/* ---------------------------------------------------------- */

int varargs(int startpos, char **results, int capacity)
{
    if (argparse_load() < 0)
        return ARGPARSE_NOTERM;

    if (startpos < 0 || startpos >= argparse_argc)
```

```
        return ARGPARSE_OOBOUND;

    int count = argparse_argc - startpos;

    /*
     * +1 because results is NULL terminated.
     */
    if (results == NULL || capacity < count + 1)
        return ARGPARSE_OOBOUND;

    for (int i = 0; i < count; i++)
        results[i] = argparse_argv[startpos + i];

    results[count] = NULL;

    return startpos;
}


/* ---------------------------------------------------------- */
/* raw argv access                                            */
/* ---------------------------------------------------------- */

int get_arg(int n, char **name, char **value)
{
    if (argparse_load() < 0)
        return ARGPARSE_NOTERM;

    if (n < 0 || n >= argparse_argc)
        return ARGPARSE_OOBOUND;

    if (name == NULL || value == NULL)
        return ARGPARSE_OOBOUND;

    char *arg = argparse_argv[n];

    *value = NULL;

    /*
     * --option
     */
    if (strncmp(arg, "--", 2) == 0) {

        *name = arg + 2;

        if (n + 1 >= argparse_argc ||
            option_like(argparse_argv[n + 1]))
            return ARGPARSE_NOVALUE;

        *value = argparse_argv[n + 1];
        return n;
    }

    /*
     * -flag / -option
     */
```

```
    if (arg[0] == '-' && arg[1] != '\0') {

        *name = arg + 1;

        /*
         * Could be a flag, therefore absence of a following
         * value is not considered an error here.
         */
        if (n + 1 < argparse_argc &&
            !option_like(argparse_argv[n + 1])) {
            *value = argparse_argv[n + 1];
        }

        return n;
    }

    /*
     * ordinary word
     */
    *name = arg;
    return n;
}
```

- Link `link_6d0edf8c-7c32-452a-ab0d-dd384f8cfffc`: node_b886239f-e4d3-490b-be5d-4e0d0792565b:out → node_98b1de57-6b1a-4a96-8e55-6e9a11eb1fc4:in (lib,)
- Link `link_1a6bae84-65aa-43b8-b8f7-3fe067f296f8`: node_98b1de57-6b1a-4a96-8e55-6e9a11eb1fc4:out → node_3b600a6f-96a5-42b4-beae-b74338ccf170:in (lib,)

argslib

- ID: `link_1a6bae84-65aa-43b8-b8f7-3fe067f296f8`
- Kind: Link
- Role: Link
- Path: ticker/argslib
- Language: c
- Technology: c-native
- Status:
- Assignee:
- Responsible: devlin

Link contract


```
{
  "sourceNodeId": "node_98b1de57-6b1a-4a96-8e55-6e9a11eb1fc4",
  "sourcePortName": "out",
  "targetNodeId": "node_3b600a6f-96a5-42b4-beae-b74338ccf170",
  "targetPortName": "in",
  "transportKind": "in-process",
  "interactionKind": "lib",
  "typeName": "",
  "payloadDefinition": "",
  "typeDefinitionId": "",
  "compositeBoundaryIds": [],
  "typeExpression": null
}
```

read_time

- ID: `node_c43e7a55-0e44-428b-972a-6dab999df396`
- Kind: Processor
- Role: Generator
- Path: ticker/read_time
- Language: c
- Technology: c-native
- Status: FUNCTIONAL
- Assignee:
- Responsible: devlin

Specification

reads system time and puts it in a Packet and then pushes it into the ip_time output

Source

```
time_t now = time(NULL);
struct tm time_info;
char timestamp[32];

// should raise a compilation error // some_nonexisten_function==!!


Packet packet = {
    .timestamp = timestamp,
    .uuid = NULL,
    .random_word = NULL,
    .random_word_length = 0
};

if (now != (time_t)-1 &&
    localtime_r(&now, &time_info) != NULL &&
    tm__strftime(
        timestamp,
        sizeof(timestamp),
        "%Y-%m-%dT%H:%M:%S%z",
        &time_info
    ) > 0) {
    packet.isok = true;
} else {
    timestamp[0] = '\0';
}



packet.timestamp = strdup(timestamp);

printf("read_time.timestamp:%s\n",packet.timestamp != NULL ? packet.timestamp : "null");
```

```
push(ip_time, &packet);
```

- Link `link_ecc22fe5-032c-4b24-812b-cc93b4ca76b0`: node_82203592-4fed-4fd4-b0bf-9b02a90b0ad0:out → node_c43e7a55-0e44-428b-972a-6dab999df396:in (lib,)
- Link `link_d1b9d286-d43a-439a-bee0-e66bf2943400`: node_c43e7a55-0e44-428b-972a-6dab999df396:out → node_627a3a3e-718f-4867-a71e-584b2d9b138b:in (data,)

uuid_lib

- ID: `node_b153ded5-bb9d-438c-9854-84fa0e737261`
- Kind: Processor
- Role: ServiceLibrary
- Path: ticker/uuid_lib
- Language: c
- Technology: c-native
- Status: IMPLEMENTATION
- Assignee:
- Responsible: devlin

Specification

```
includes the necessary headers to generate uuids is gona be used by various functions
```

Source

```
#ifndef UUID_LIB_H
#define UUID_LIB_H

#include <uuid/uuid.h>
#include <stdbool.h>
#endif /* UUID_LIB_H */
```

- Link `link_2f93c7bf-0dcc-4589-8e80-ad66d57b3862`: node_b886239f-e4d3-490b-be5d-4e0d0792565b:out → node_b153ded5-bb9d-438c-9854-84fa0e737261:in (lib,)
- Link `link_a17dca3c-a162-4f7f-a264-abcd20c1f421`: node_b153ded5-bb9d-438c-9854-84fa0e737261:out → node_627a3a3e-718f-4867-a71e-584b2d9b138b:in (lib,)

write_append_file

- ID: `node_3b600a6f-96a5-42b4-beae-b74338ccf170`
- Kind: Processor
- Role: Sink
- Path: ticker/write_append_file
- Language: c
- Technology: c-native
- Status: FUNCTIONAL
- Assignee:
- Responsible: devlin

Specification

takes a Packet from ip_word, serializes it as a CSV record, and appends the result to "ticker_out.csv"

Source

```
Packet item;

char *value;
int sleepval=5000;
int p = argslib__option("sleep", &value);
if(p>=0){
    sleepval=atoi(value);
}
printf("write_append_file.sleepvalue:%d\n",sleepval);


if (pop(ip_word, &item) != THREADWORK_OK) {
    return THREADWORK_OK;
}

FILE *file = fopen("ticker_out.csv", "a");
if (file == NULL) {
    return THREADWORK_ERROR;
}

int fprintf_result = fprintf(
    file,
    "\"%s\",\"%s\",\"%s\",%.0f,%s\n",
    item.timestamp != NULL ? item.timestamp : "",
    item.uuid != NULL ? item.uuid : "",
    item.random_word != NULL ? item.random_word : "",
    item.random_word_length,
    item.isok ? "true" : "false"
);
printf("write_append_file.timestamp:%s\n",item.timestamp != NULL ? item.timestamp : "null");
printf("write_append_file.uuid:%s\n",item.uuid != NULL ? item.uuid : "null");
printf("write_append_file.random_word:%s\n",item.random_word != NULL ? item.random_word : "null");

if (fprintf_result < 0) {
    fclose(file);
    return THREADWORK_ERROR;
}

if (fclose(file) != 0) {
    return THREADWORK_ERROR;
}

free(item.timestamp);

printf("write_append_file.sleeping 5000 ns \n");
usleep(sleepval);
```

- Link `link_1a6bae84-65aa-43b8-b8f7-3fe067f296f8`: node_98b1de57-6b1a-4a96-8e55-6e9a11eb1fc4:out → node_3b60

0a6f-96a5-42b4-beae-b74338ccf170:in (lib,)
- Link `link_3f327a51-135b-4216-b55f-70ae41f06a30`: node_b5041d57-c1e9-4f67-9e88-760c35c8efde:out → node_3b60
0a6f-96a5-42b4-beae-b74338ccf170:in (lib,)
- Link `link_dd5fcd39-8d54-4310-b6fe-242d927f0ec5`: node_059814d5-2184-47b6-93b0-f754589cf34f:out → node_3b60
0a6f-96a5-42b4-beae-b74338ccf170:in (data,)

README.md

- ID: `node_5b6c66ab-3d46-41fb-96ff-dee9c1738292`
- Kind: Processor
- Role: ProcessingUnit
- Path: ticker/README.md
- Language: shellscript
- Technology: file-export
- Status: DEPLOYED
- Assignee:
- Responsible: devlin

Source

```
this is something that i wrote
```

96.png

- ID: `node_4b032846-a445-4327-95db-4178aa447304`
- Kind: Processor
- Role: ProcessingUnit
- Path: ticker/96.png
- Language: plain
- Technology: file-export
- Status: DEPLOYED
- Assignee:
- Responsible: devlin

qjs.dll

- ID: `node_078ec47e-e735-4cc2-a56a-34da3933ac8b`
- Kind: Processor
- Role: ProcessingUnit
- Path: ticker/qjs.dll
- Language: plain
- Technology: file-export
- Status: DEPLOYED
- Assignee:
- Responsible: devlin

qjs.dylib

- ID: `node_352d6efc-bce0-4273-8c4b-03c64e7764fc`
- Kind: Processor
- Role: ProcessingUnit
- Path: ticker/qjs.dylib
- Language: plain
- Technology: file-export
- Status: DEPLOYED

- Assignee:
- Responsible: devlin

qjs.so

- ID: `node_6a831982-71aa-47ad-ab14-fbfcbc9cd384`
- Kind: Processor
- Role: ProcessingUnit
- Path: ticker/qjs.so
- Language: plain
- Technology: file-export
- Status: DEPLOYED
- Assignee:
- Responsible: devlin

Signal

- ID: `node_34e10e8d-c693-4e56-91d9-f03334aafa9b`
- Kind: Type
- Role: Type
- Path: ticker/Signal
- Language: c
- Technology: c-native
- Status: DEPLOYED
- Assignee:
- Responsible: devlin

Packets

- ID: `node_968adf46-fe80-45e7-a6c3-90d78a8697e4`
- Kind: Type
- Role: Type
- Path: ticker/Packets
- Language: c
- Technology: c-native
- Status: BUSINESS
- Assignee:
- Responsible: devlin